Project #16: In-Situ Moisture Sensing of Timber Structures

Final Year Project Report

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such as Marie Patlong

Abstract

Moisture is a leading cause of physical and biological degradation in timber, yet the pin-type meters used in practice are invasive and give only localised, intermittent readings. This project developed and tested a low-cost, non-destructive, in-situ moisture sensing system for New Zealand timber using Laser-Induced Graphene (LIG) interdigital capacitive sensors. Four five-finger-pair sensors were engraved directly onto borax-treated Radiata Pine using a blue diode laser with a wavelength of 450 nm. Capacitance was measured with a Texas Instruments FDC1004 capacitance-to-digital converter and a Seeed Studio XIAO ESP32-C3, which streamed readings over Wi-Fi to a custom Python application for real-time plotting and CSV logging. Each sensor was calibrated individually with a linear regression, giving R^2 values of 0.23 to 0.69. In an 8-hour drying trial following surface wetting, two sensors showed the expected sharp capacitance peak followed by a smooth decay, while the other two behaved anomalously, which is attributed to degradation of unencapsulated graphene by direct water contact. The results show that the sensing principle and the integrated acquisition pipeline work, but that quantitative moisture estimates are not yet reliable and that field use will require protective, vapour-permeable encapsulation. Future work should expand the calibration dataset, validate readings against an independent moisture reference, and extend testing to longer cycles and bonded mass timber elements.

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Glossary of Terms

LIG Laser-Induced Graphene: A porous, conductive carbon material produced by localized laser pyrolyzing of carbonaceous substrates such as timber.

MC Moisture Content: The mass of water present in a timber sample expressed as a percentage of its oven-dried mass.

FSP Fiber Saturation Point: The moisture threshold (typically 28%–30%) where cell walls are fully saturated with bound water.

IDT Interdigital Transducer: A comb-like coplanar electrode pattern used to generate fringing electric fields for capacitive sensing.

CDC Capacitance-to-Digital Converter: An integrated circuit (e.g., TI FDC1004) that converts small capacitance changes into digital signals.

CAPDAC Capacitance Offset DAC: Internal circuitry on the FDC1004 used to cancel baseline parasitic capacitance up to 100 pF.

1 Introduction

The adoption of high-value wood and mass timber products is a cornerstone of New Zealand's strategy for a low-emissions construction sector [1]. However, the hygroscopic nature of timber requires precise, real-time moisture monitoring to prevent physical and biological degradation. This project builds upon prior foundational research [2], which successfully demonstrated the feasibility of Laser-Induced Graphene (LIG) sensors on both Kapton tape and solid wood substrates. While the previous study established the validity of these sensors, they are yet to be implemented in a fully integrated, in-situ monitoring system. This project aims to integrate LIG sensors with an ESP32-C3 embedded system and a separate custom PC program for real-time data trending. This technology could strengthen Timber's confidence as a construction material in Aotearoa, New Zealand.

1.1 Scope of Research

The scope of this project is to develop and validate a low-cost, non-destructive moisture sensing system for New Zealand timber. The project focuses on the intersection of material science, sensing, and monitoring.

In-Scope:

- **Fabrication:** Design and laser-fabrication of interdigitated capacitive sensors using Laser-Induced Graphene (LIG) directly on Radiata Pine substrates.
- **Hardware Development:** Design of a signal conditioning circuit and integration with an ESP32-C3 microcontroller to measure small changes in capacitance.
- **Firmware:** Development of a C++ based real-time data acquisition system with a sampling rate suitable for tracking moisture fluctuations.
- **Software:** A Python-based program for remote data visualisation, real-time trending, and CSV-based data logging.
- **Validation:** Laboratory-based proof-of-concept testing using solid sawn Radiata Pine to simulate the moisture behaviour found in both traditional framing and individual mass timber lamellas.

Out of Scope (Exclusions):

- **Alternative Sensing Modes:** This project will not investigate resistive or power-loss sensing methods.
- **Full-Scale Implementation:** The system will not be installed in an active Mass Timber building site or within bonded Cross Laminated Timber (CLT) panels.

- **Environmental Hardening:** High-level waterproofing or industrial-grade enclosures for the electronics are not within the current scope.

1.2 Project Objectives

To address the identified knowledge gap regarding low-cost, in-situ monitoring in the NZ timber industry, the following objectives have been established:

1. **Material:** Characterize the electrical response of LIG-on-wood sensors across a moisture range of 8% to 30% (Fiber Saturation Point).
2. **Electronics:** Develop a functional embedded system using the ESP32-C3 that can accurately convert analogue sensor signals into calibrated Moisture Content (MC%) values.
3. **Data:** Implement a wireless data transmission protocol to a Python-based PC application for real-time monitoring and historical data storage.
4. **Verification:** Verify the accuracy of the LIG-capacitive system.

2 Literature Review

2.1 Timber's Increasing Significance

Timber is becoming increasingly important in modern construction, evolving from solid sawn wood used in building frames since antiquity to modern engineered variants such as Mass Timber developed in the early 1990s. Examples include Cross-Laminated Timber (CLT), Nail-Laminated Timber (NLT), Glue-Laminated Timber (Glulam), and Dowel-Laminated Timber (DLT) [3]. Timber presents a sustainable alternative to traditional concrete and steel; however, its widespread adoption remains constrained by fire safety and long-term durability concerns.

The New Zealand Government is actively promoting onshore timber processing capabilities [1]. An increased demand for mass timber products within the domestic construction sector will directly reinforce this policy framework.

2.2 Material Properties and Moisture Mechanics of Timber

Moisture Content (MC) is quantitatively defined as the mass of water expressed as a percentage of the oven-dried mass of the wood [3]:

$$\mathrm{MC} = \frac{m_{\mathrm{water}}}{m_{\mathrm{dry}}} \times 100\% = \frac{m_{\mathrm{sample}} - m_{\mathrm{dry}}}{m_{\mathrm{dry}}} \times 100\% \tag{1}$$

As a naturally hygroscopic, organic polymer, wood is inherently susceptible to moisture-induced

physical and biological degradation [3]. Fluctuations in MC can trigger two primary modes of structural failure:

1. **Physical Degradation:** Moisture gradients across wood fibers generate localized internal stresses [4]. These stresses lead to dynamic structural warping—manifesting as localized expansion, crushing, severe micro-cracking in sawn timber, and bondline delamination in mass timber element layers.
2. **Biological Degradation:** Sustained elevated MC levels foster ideal growth environments for fungal decay, bacterial action, and insect attack. The threshold between 28% and 30% MC defines the Fiber Saturation Point (FSP), where cell wall bound water transitions to free liquid water within cellular cavities. Exceeding the FSP drastically increases the rate of biological and fungal decay [3].

2.3 Current Practice

The oven-dry method remains the ”gold standard” for determining Moisture Content (MC) in a laboratory setting; however, it is destructive, time-consuming, and unsuitable for field deployment [3]. In field applications, the indirect conductance method using pin-type sensors represents the current state of the art. These sensors determine MC by measuring electrical resistance between metal probes driven into the timber. Despite their prevalence, pin-type sensors exhibit notable limitations:

- **Invasiveness:** Penetrating the timber surface causes physical damage to structural members and protective weather membranes.
- **Localized Snapshots:** Measurements are highly localized to the pin insertion point and fail to provide a holistic view of structural health or the continuous logging required for long-term monitoring.
- **Measurement Uncertainty:** Readings are significantly influenced by insertion depth, local wood density variations, and ambient temperature fluctuations.

To overcome these constraints in large-scale structural health assessments, researchers have utilized the hygrometric method as a non-invasive alternative for long-term monitoring. As detailed by Imamura et al. [4] in their study on Cross-Laminated Timber (CLT) basement walls, this approach involves machining internal cavities into the timber at specific depths (e.g., 2.5 cm and 6.35 cm) to evaluate individual plies. Relative Humidity (RH) and temperature sensors are inserted into these cavities and sealed with silicone sealant to ensure moisture exchange occurs solely within the isolated void air space. Rather than evaluating direct electrical properties, the MC is estimated as the Equilibrium Moisture Content (EMC) via empirical formulas derived from internal RH and temperature readings [4].

While the hygrometric approach effectively captures structural swelling and shrinking phases for strain correlation, it relies on discrete commercial sensors and extensive external wiring networks. Furthermore, field deployments suffer from reliability challenges; Imamura et al. noted that data-acquisition instabilities can take months to resolve and sensors are frequently damaged during construction activities such as backfilling [4]. This highlights a critical industry gap for an integrated "sensor-as-structure" monitoring framework using Laser-Induced Graphene (LIG).

2.4 Other Emerging Technologies

Recent advancements have introduced Lead Zirconate Titanate (PZT) patches for active stress wave sensing in timber structures [5]. In this signal-centric setup, one PZT transducer acts as an actuator to generate stress waves through the timber body while a paired PZT transducer acts as a receiver:

- **Physics of Attenuation:** Variations in MC alter both the density and elastic modulus of wood. As increases, the propagation velocity and signal amplitude of the stress wave decrease, providing a metric for moisture detection [5].
- **Machine Learning Interpretation:** Researchers employ Naïve Bayesian classification models to process complex wave attenuation patterns, achieving accurate MC classification even in the presence of structural timber cracks [5].
- **Comparative Drawbacks:** Although accurate, PZT active sensing systems require bulky hardware, high-cost signal generators, and intensive signal processing, limiting their viability for scalable field installation [5]. This creates a clear opportunity for lightweight, low-cost integrated solutions like LIG.

2.5 The System: Capacitive vs. Resistive LIG Sensing

LIG offers a material-centric alternative to conventional moisture sensors, as it can be fabricated directly onto the timber surface. While most LIG-based moisture sensors operate chemiresistively, relying on absorbed water molecules within the porous graphene network to alter charge-carrier transport, this project investigates capacitive sensing to detect changes in the dielectric properties of wood.

The dielectric permittivity of timber increases with moisture content (MC) due to the high permittivity of water relative to dry wood. Consequently, capacitive measurements can be used to monitor changes in timber moisture content. By patterning LIG into interdigitated electrodes (IDEs), the sensor can detect variations in the effective dielectric permittivity of the surrounding material.

To enable high-resolution capacitance measurements, the sensor interfaces with the Texas Instruments FDC1004 capacitance-to-digital converter (CDC), which provides a nominal capacitance

resolution of 0.5 fF [6]. Its integrated capacitance-to-digital offset DAC (CAPDAC) compensates for large baseline capacitances of up to 100 pF, helping accommodate the parasitic capacitance associated with sensor wiring and larger-scale timber structures [6].

Software and System Integration: A Seeed XIAO ESP32-C3 microcontroller acquires capacitance measurements from the FDC1004 via the I^2C bus and transmits the data wirelessly to a custom Python-based graphical user interface (GUI). The application supports real-time visualisation, time-window selection, data filtering, and persistent logging, enabling continuous monitoring in place of manual, point-in-time measurements.

An additional advantage of capacitive sensing is its potential for reduced temperature sensitivity compared with resistive moisture sensing [7]. Electrical resistance in wood varies with both moisture content and temperature, complicating the interpretation of resistance measurements and potentially necessitating temperature compensation and multivariable calibration. Capacitive measurements are also influenced by temperature, but may exhibit greater stability under appropriate operating conditions [8]. This makes capacitive sensing a promising approach for long-term timber moisture monitoring, although its temperature dependence must still be characterised for the intended application.

3 Methods and System Design

3.1 Sensor Fabrication

Laser-Induced Graphene (LIG) moisture sensors were fabricated directly onto timber substrates to produce low-cost capacitive moisture sensors. The substrates were Radiata Pine strips. Prior to laser processing, timber samples were immersed in a borax solution (saturated solution) overnight before being allowed to dry under ambient conditions. The borax treatment promotes graphitisation during laser engraving and improves the quality and consistency of the resulting graphene electrodes.

Following treatment, an Interdigital Transducer (IDT) pattern was engraved directly onto the timber surface using a blue diode laser on a Snapmaker machine with a Blue diode laser, 450 nm and a rated maximum power of 1600 mW), with the settings listed in Table 1. The laser converts the treated surface into porous conductive graphene while simultaneously defining the electrode geometry.

Electrical connections were established by attaching header pins to each sensor, allowing connection to the capacitance measurement electronics. Four nominally identical sensors were fabricated for simultaneous monitoring.

Table 1: Laser engraving parameters used to fabricate the LIG electrodes.

Parameter	**Value**
Method	Fill (line-filled engraving)
Movement mode	Line
Line direction	Horizontal
Fill interval	0.05 mm
Work speed	750 mm/min
Laser power	20% of rated output 1600 mW
Number of passes	1
Laser type / wavelength	Blue diode laser, 450 nm



(a) Borax treatment of timber samples.



(b) Fabricated Laser-Induced Graphene IDT sensor.

Figure 1: Fabrication process and completed LIG interdigital sensor on timber.

3.2 Sensor Design & Operating Principle

The fabricated sensors utilise an Interdigital Transducer (IDT) geometry consisting of two interlocking comb-like electrodes. Unlike parallel-plate capacitors, the electric field generated by an IDT extends into the surrounding medium via fringing fields above and below the substrate surface.

As the moisture content of the timber changes, the effective relative permittivity of the region surrounding the electrodes alters accordingly. Because water possesses a significantly higher relative permittivity ($\epsilon_r \approx 80$) than dry timber ($\epsilon_r = 2 + 0.5$ as a conservative estimate [9]), an increase in moisture content yields a corresponding increase in measured capacitance. Measuring capacitance therefore provides an indirect assessment of timber moisture content. Sensor sensitivity is primarily governed by electrode geometry, including finger width, finger spacing, and overlap length.

Multiple candidate geometries with differing numbers of finger pairs were designed and fabricated. The number of finger pairs was increased until the sensor capacitance fell within the measurement range of the capacitance-to-digital converter (approximately 10–100 pF). The five-finger-pair geometry shown in Figure 2 was ultimately selected, with a baseline dry capacitance of approximately 10–12 pF. Its dimensions are listed in Table 2.

Table 2: Geometry of the selected IDT design.

Parameter	**Symbol**	**Value (mm)**
Number of finger pairs	N	5
Finger width	w	1
Finger spacing	s	0.5
Overlap length	l	5
Overall electrode length	L	25

The theoretical capacitance per finger pair of the IDT structure can be estimated for electrode geometries satisfying the condition

$$0 < \frac{s}{s+2w} \leq \frac{1}{\sqrt{2}}, \tag{2}$$

using Hilberg's explicit formulation [10]:

$$C = \frac{\epsilon_r l \ln\left(-\frac{2}{\sqrt[4]{1-\frac{s^2}{(s+2w)^2}}-1} \left(\sqrt[4]{1-\frac{s^2}{(s+2w)^2}} + 1 \right) \right)}{377\pi v_0}, \tag{3}$$

where s is the finger spacing, w is the finger width, l is the electrode overlap length, ϵ_r is the relative permittivity of the medium, and v_0 is the speed of light in vacuum.

Razafindratsima et al. [9] experimentally determined the dielectric properties of various timber species across a broad range of moisture levels. Although their study evaluated Maritime Pine (*Pinus pinaster*), a close relative of Radiata Pine (*Pinus radiata*), the measured permittivity values for dry wood overlapped across all species tested within experimental error. For electric fields aligned parallel and perpendicular to the fibre direction, they reported relative permittivities of $\epsilon_r = 2 \pm 0.5$ and $\epsilon_r = 1.7 \pm 0.1$, respectively. Adopting $\epsilon_r = 2 \pm 0.5$ effectively models the oven-dry baseline across species, fibre orientations, and minor density variations.

For the selected sensor design (Figure 2), the geometry yields $\frac{s}{s+2w} = 0.2$, which satisfies the validity condition in Equation 2. Substituting $\epsilon_r = 2 \pm 0.5$ into Equation 3 yields a theoretical capacitance of 0.168 ± 0.04 pF per finger pair. The selected sensor has five finger pairs, giving a total theoretical capacitance of approximately 0.841 ± 0.2 pF. This is lower than the 10–12 pF measured experimentally from oven-dried timber samples. The difference is likely attributable to stray capacitance within the measurement setup (such as the long jumper wires, header pins and connection tracks), together with residual moisture within the wood matrix. The full working for finding out the theoretical capacitance can be found in appendix A

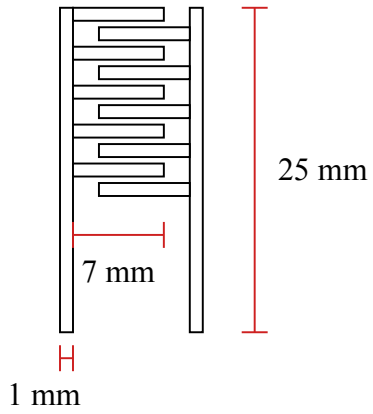


Figure 2: IDT geometry.

3.3 Measurement Hardware

Capacitance measurements are performed using a Texas Instruments FDC1004 Capacitance-to-Digital Converter connected to a Seeed Studio XIAO ESP32-C3 microcontroller.

The FDC1004 provides four independent measurement channels, allowing four moisture sensors to be monitored simultaneously. It measures capacitance directly over a limited range, and a programmable offset (CAPDAC), adjustable up to approximately 100 pF, subtracts the baseline capacitance so that larger sensor capacitances can be measured.

Communication between the ESP32-C3 and FDC1004 is achieved using the I^2C protocol. The

ESP32 executes firmware written in C++ that periodically samples each sensor at 100 Hz and transmits the measured capacitance values wirelessly to a host computer over Wi-Fi via WebSockets. 100 Hz is a much higher rate than necessary for measuring changing moisture levels in timber, much faster than the natural fluctuations in timber MC but is the slowest sampling rate available on the FDC1004, to compensate for this the host computer implements down-sampling, averaging over a window of an adjustable size. Figure 3 illustrates the overall hardware architecture.

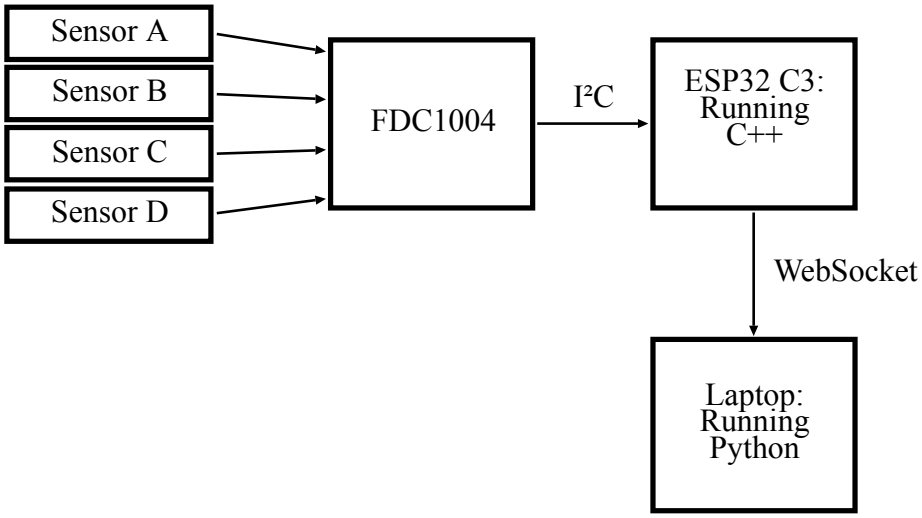


Figure 3: Hardware block diagram displaying sensor interface to FDC1004 and ESP32-C3.

3.4 Data Acquisition Software

A custom data acquisition software platform, named `Timber_moisture_content_analyser`, was developed to receive, store, and visualise moisture sensor measurements in real time.

The ESP32 transmits capacitance measurements to a Python application via a WebSocket connection over a local wireless network. The Python software architecture (Figure 5) consists of four primary modules:

- **Receiver:** Continuously listens for incoming sensor measurements over the network stream.
- **Logger:** Records incoming timestamps and raw capacitance values directly to a local CSV file for persistent storage. Implementing down-sampling over a window of an adjustable size.
- **Live Plotter:** Displays live, multi-channel graphical trendlines from all four sensors during experimental runs.
- **Historical Plotter:** Loads recorded historical CSV datasets to inspect specific time windows of interest.

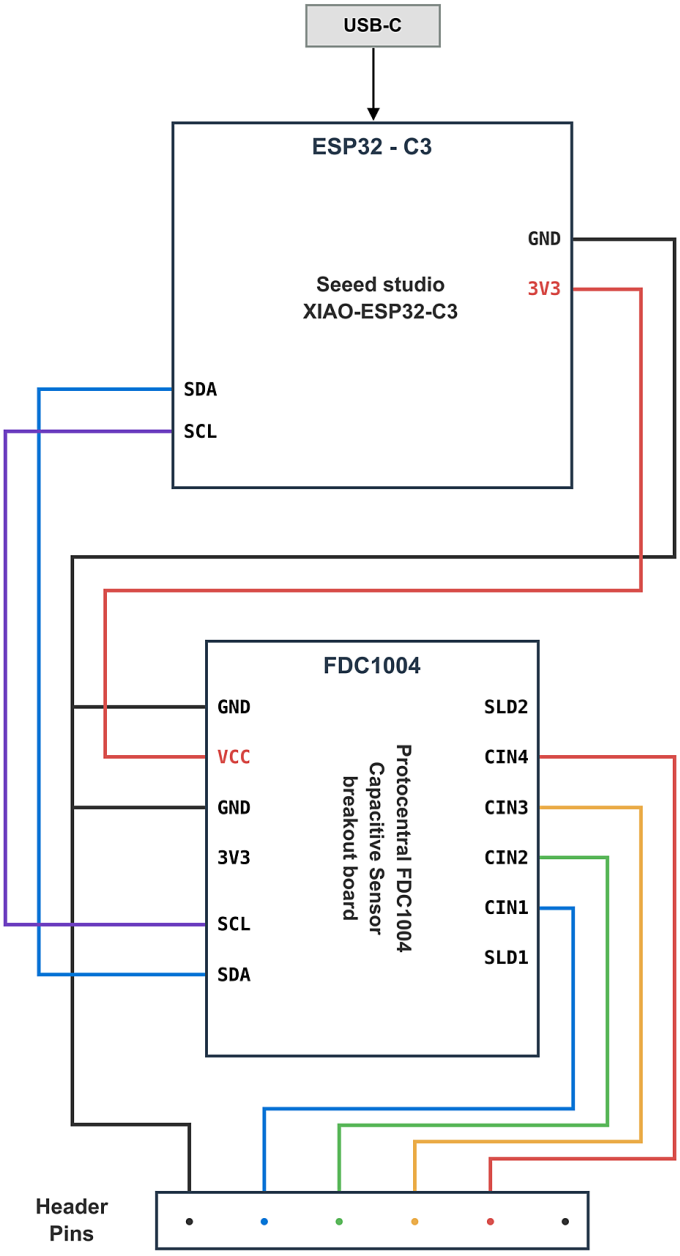


Figure 4: Electrical wiring diagram showing the ESP32-C3, FDC1004 breakout board and sensor header pins.

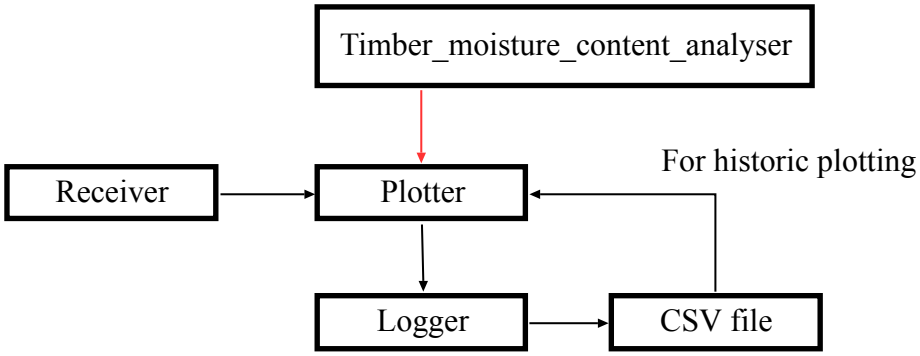


Figure 5: Software architecture and data processing pipeline.

4 Calibration and Preliminary Results

4.1 Determining Known Moisture Contents for Calibration

Calibrating each sensor required a reference measurement of timber moisture content. For this project, a commercial off-the-shelf resistive pin-type moisture meter was used: the **POROMETISTO WTS01Pro Pin & Pinless Moisture Meter.** The manufacturer states a measuring range of 6–60% MC for wood and a resolution of 0.1%. The meter was used exclusively in pin mode and in its wood setting, which is intended for timber, rather than its brick setting, which is intended for masonry. The meter was allowed to calibrate before each session by pressing the "MEAS" button, where the device made a beep and "Cal" came up on its display A typical meter display is shown in Figure 6.

Readings were taken by lightly pressing the pins into the sample (figure 7) so that only the pin tips penetrated the surface, to prevent excessive damage to the sample, then allowing the reading to settle to a roughly constant value over a few seconds (Figure 8). Because the pin tips penetrate only the surface, each reading reflects near-surface moisture content, which may differ from the bulk value in samples that have not reached moisture equilibrium, but that is more related to the permittivity that the IDT is actually measuring. The stated precision of the meter also sets a lower bound on the uncertainty of the reference values used for calibration. The oven-dry reference point (0% MC) was determined by fully drying the sample at 60 °C in a laboratory drying oven overnight and the WTS01Pro agreed, giving a 0% MC measurement despite outside the manufacturers measuring range, the samples where given 1 minute to cool before capacitance measurement were made.



Figure 6: POROMETISTO WTS01Pro in Pinned mode with typical display

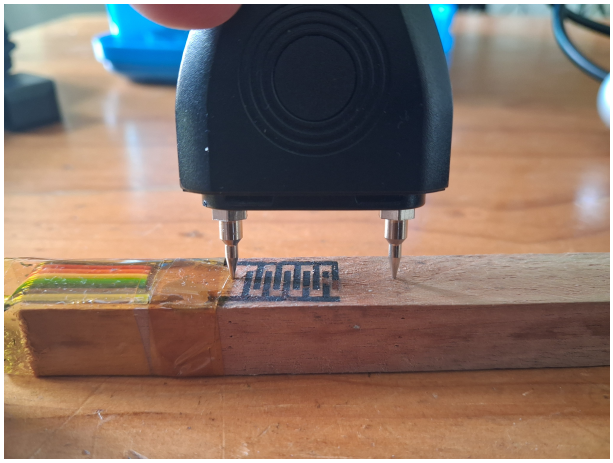


Figure 7: How pins were applied to sample for measurement



Figure 8: settled reading and sensor

4.2 Sensor Calibration

Each sensor was calibrated by measuring capacitance across a range of known timber moisture contents. Calibration was performed individually for each sensor to account for manufacturing variations inherent to the laser fabrication process. Reference moisture contents were determined as described in 4.1 and each sensor was calibrated with 10 points.

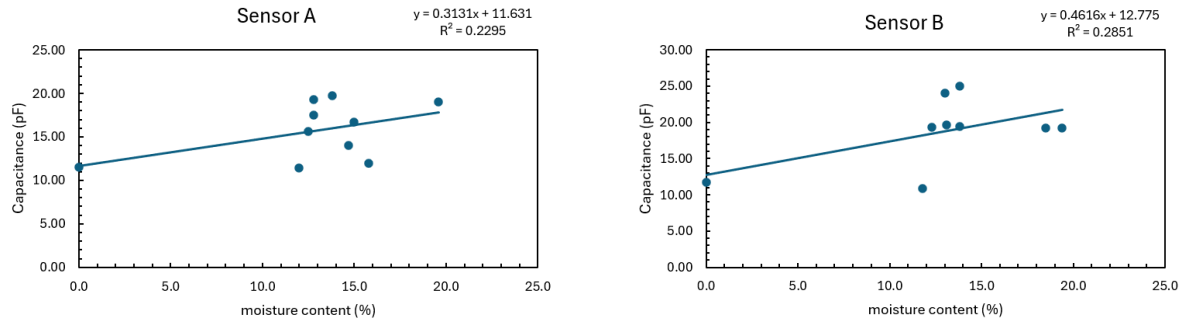
Calibration points consisted of an oven-dry reference (0% MC) and a set of conditioned samples between approximately 11% and 20% MC. The lower end of this range corresponds to the equilibrium moisture content of the timber at ambient laboratory humidity, below which samples could not be dried without oven drying. Above approximately 20% MC, liquid water bridged the interdigital fingers and produced unreliable readings, preventing calibration towards the fibre saturation point. The calibration therefore does not cover the intermediate range between 0% and 11% MC, or the range above 20% MC.

Measured capacitance values (C_i) were converted to estimated Moisture Content ($MC_i\%$) using linear regression, with moisture content as the dependent variable:

$$MC_A = 0.7330C_A + 1.4138 \tag{4}$$

$$MC_B = 0.6177C_B + 1.2982 \tag{5}$$

where MC represents the moisture content (%) and C represents the measured capacitance in picofarads (pF). Figure 9 plots capacitance against moisture content for visual clarity, so the fitted lines shown there are the regression of C on MC and differ from the equations above, which regress MC on C.



(a) Sensor A ($y = 0.3131x + 11.631$, $R^2 = 0.2295$)

(b) Sensor B ($y = 0.4616x + 12.775$, $R^2 = 0.2851$)

Figure 9: Experimental calibration curves plotting capacitance (pF) against moisture content (%) for Sensors A and B.

4.3 Preliminary Experimental Results and Physical Stability

Initial testing showed that all four fabricated LIG sensors responded to changes in timber moisture content. The calibration data were approximately linear in form but showed considerable scatter ($R^2 = 0.23$ to 0.69), and sensitivity differed between nominally identical sensors, from 0.31 to 0.84 pF per % MC.

Repeated wetting of the sensors resulted in visible degradation of the laser-induced graphene surface (Figure 10). This reduced the raw sensor response. Adjusting the y-intercept with a drift coefficient as:

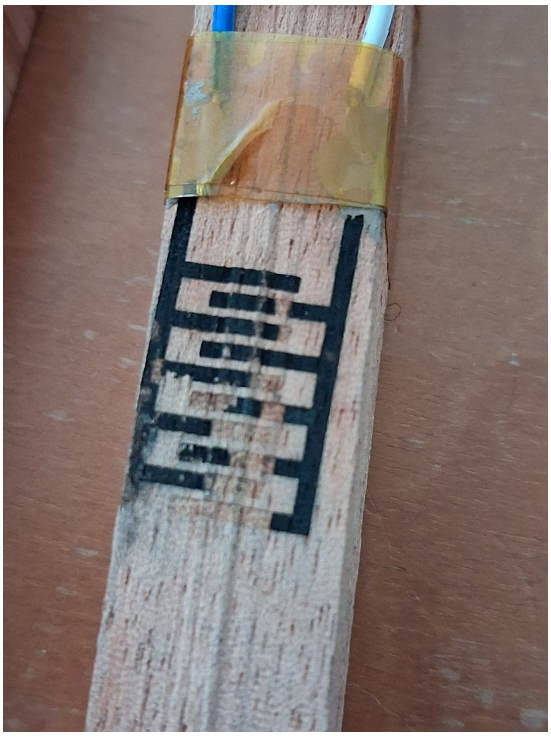
$$\text{Drift Coefficient A} = +3.7\ \%$$

$$\text{Drift Coefficient B} = +2.9\ \%$$

partially compensated for this reduction, although an intercept correction cannot fully account for a loss of sensitivity.



(a) Sensor without water damage.



(b) Sensor exhibiting physical water degradation.

Figure 10: Visual comparison of different LIG sensor traces and how water exposure has effected them differently.

4.4 System Integration Overview

The completed measurement system combines the fabricated LIG sensors, capacitance measurement electronics, and custom software into a complete moisture monitoring platform. The integrated system provides:

- Simultaneous monitoring across four independent channels.
- Wireless data transmission over Wi-Fi.
- Real-time moisture estimation via embedded/software calibration formulas.
- Automated CSV data logging and time-stamping.
- Live and historical graphical plotting capabilities.

5 Results

5.1 Dynamic Moisture Monitoring and System Response

To evaluate the real-time performance of the integrated sensing platform, continuous time-series monitoring of all four channels was conducted over an 8-hour period following a controlled

surface wetting event at $t = 0$ hours. All sensors began from a timber moisture content of approximately 10% MC, as measured with the pin-type reference meter (Section 4.1).

Two of the four sensors showed anomalous behaviour, attributed to damage to the graphene from direct water contact, and are excluded from the main results. The remaining two sensors are reported here as Sensor A and Sensor B, corresponding to logged channels CIN2 and CIN4 respectively (Table 3 in Appendix B). The excluded sensors were identified as anomalous because one sensor read approximately 5.7% MC at the start of the trial against the 10% reference and responded only weakly to wetting, and the other showed a collapse in its reading between $t = 0.5$ and $t = 1.2$ hours that is not consistent with drying. Their traces are given in Appendix B.

Reporting two of four sensors limits the evidence for sensor performance. Surprisingly the sensors that were retained also have the weaker calibration fits ($R^2 = 0.23$ and 0.29, Figure 9). This suggests the possibility of some overfitting. At $t = 0$ the retained sensors read approximately 9.8% and 10.3% MC, within about 0.3 percentage points of the 10% reference.

Figure 11 shows the calculated moisture content of the two retained sensors, and Figure 12 shows each sensor's moisture content alongside its raw capacitance (pF).

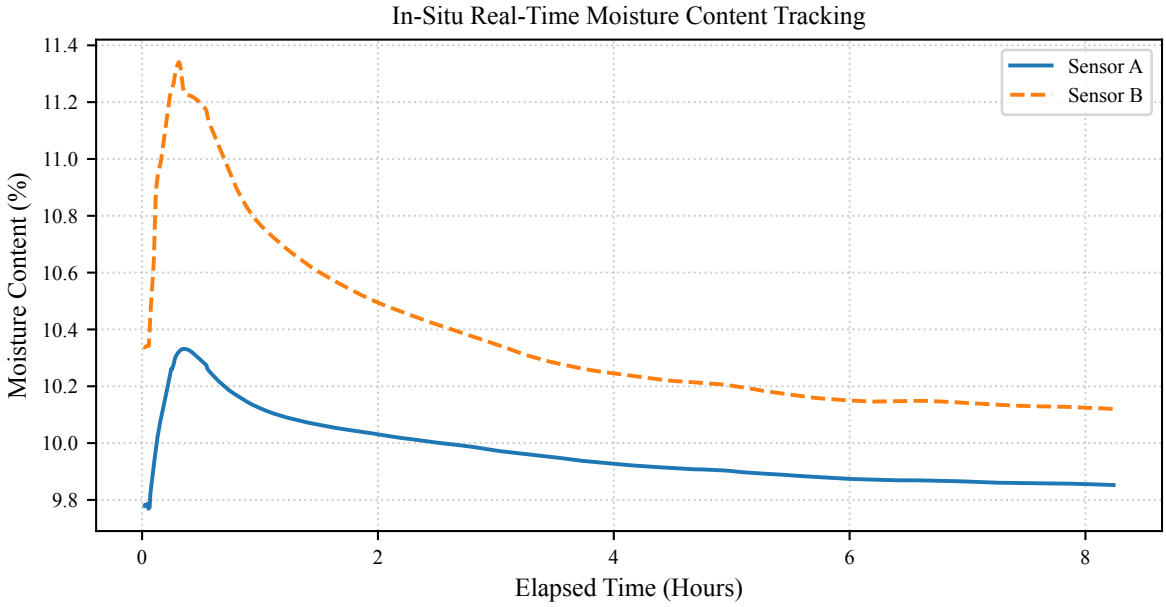


Figure 11: Real-time continuous moisture tracking for Sensors A and B over an 8-hour drying cycle following initial surface wetting at $t = 0$.

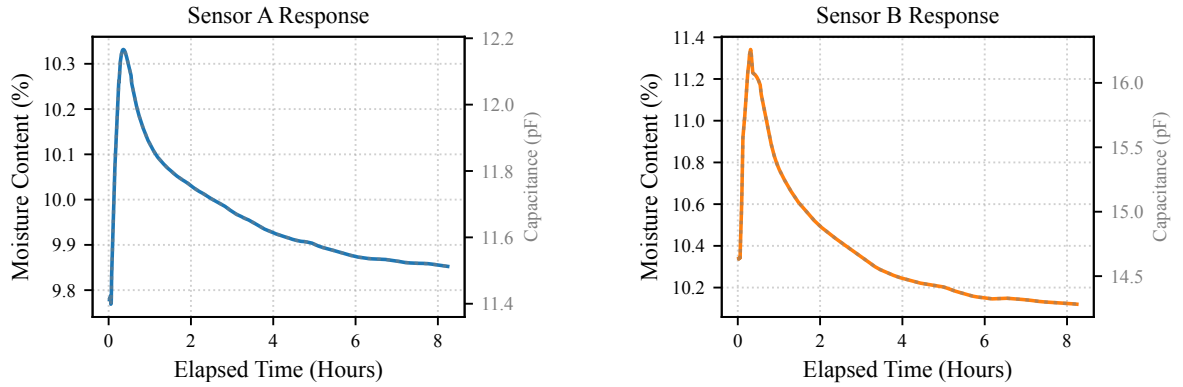
5.2 Channel Performance and Response Dynamics

Both retained sensors responded to wetting and drying in the expected way:

- **Transient peak ($t = 0$ to $t = 0.3$ hrs):** Upon water application, both sensors showed

an immediate rise in measured capacitance, reaching peak estimated moisture contents of 10.3% (Sensor A) and 11.3% (Sensor B) within about 20 minutes. These peaks are rises of roughly 0.5 and 1.0 percentage points above the starting readings.

- **Desorption decay ($t = 0.3$ to $t = 8$ hrs):** After the peak, both channels decayed smoothly as moisture evaporated into the ambient environment, approaching equilibrium values near 9.8% and 10.1% MC respectively, close to the 10% reference measured at the start.



(a) Sensor A response.

(b) Sensor B response.

Figure 12: Individual channel time-series responses of the retained sensors following initial surface wetting at $t = 0$.

5.3 Physical Stability and Limitations of Unencapsulated LIG

Two of the four fabricated sensors showed anomalous behaviour attributable to physical graphene damage from direct liquid exposure (Appendix B):

- **Excluded sensor 1** recorded minimal sensitivity, possibly due to reduced electrical contact integrity at the header pin interface during wetting.
- **Excluded sensor 2** showed a severe transient drop between $t = 0.5$ and $t = 1.2$ hours before recovering. This is consistent with temporary liquid bridging across the interdigital fingers, which would disturb the fringing field until surface moisture dropped below saturation.

Neither mechanism was confirmed. That half of the sensors were affected shows that the unencapsulated LIG traces are vulnerable to liquid water. Practical applications are therefore likely to require protective, vapour-permeable encapsulation to prevent mechanical and liquid-induced degradation of the carbon matrix.

6 Discussion

The results show that Laser-Induced Graphene (LIG) interdigital capacitive sensors on Radiata Pine respond to changes in timber moisture, and that the integrated ESP32-C3-to-PC pipeline can log and display that response wirelessly through an 8-hour continuous drying event without manual intervention. Only two of the four fabricated sensors are reported in the main results, so this section also considers what can and cannot be concluded from that subset. It examines the relationship between the calibration and dynamic results, the physical limitations of the unencapsulated sensors, and the limitations of the study.

6.1 Reconciling Calibration Fit and Dynamic Response

The calibration results (Figure 9) and the dynamic response data (Figures 11 and 12) do not tell a consistent story. Sensors A and B have the two weakest linear calibration fits ($R^2 = 0.23$ and 0.29), yet showed the behaviour expected during the drying cycle: a sharp transient peak following wetting, then a smooth decay as the surface dried. The two excluded sensors calibrated more strongly ($R^2 = 0.67$ and 0.69) but showed anomalous responses (Appendix B). The goodness of fit of the calibration therefore did not predict how well a sensor behaved dynamically.

Three factors probably contribute. First, each sensor was calibrated with only a small number of points. Second, each regression includes a single oven-dry point at approximately 0% MC that is separated from the remaining cluster at roughly 11–20% MC. A single high-leverage point of this kind can raise or lower R^2 substantially without saying anything about how well the sensor performs across the range of interest. Third, a steady-state calibration and a transient drying response probe different behaviours, so a high R^2 does not imply faithful dynamic tracking, and a low R^2 does not imply a poor dynamic response. The higher R^2 values of the excluded sensors may also partly reflect fitting around damage-induced offsets rather than genuine sensitivity.

It is important to be clear about what the smooth decay in Sensors A and B does and does not show. It shows that the sensors respond in a physically plausible way to surface wetting followed by drying. It does not show that the absolute MC values are correct. During the trial the two sensors read between about 9.8% and 11.3% MC, which is at or slightly below the lowest wet calibration points (approximately 11%, the lowest level that could be conditioned at ambient humidity), so their readings depend on the regression lines at the edge of, or just outside, the calibrated range. The peak rises of roughly 0.5 and 1.0 percentage points are also small compared with the scatter in the calibration data, although the residual error was not quantified.

Sensors A and B started within about 0.3 percentage points of the 10% MC measured with the pin-type reference meter, and settled close to it by the end of the trial (9.8% and 10.1%). The excluded sensors, by contrast, read about 5.9% and 6.7% at the end of the trial, well below the reference. The agreement of the retained sensors supports their plausibility, but it is not

independent confirmation, because the sensors were calibrated against the same meter and that meter has no manufacturer-stated accuracy. Sensitivity also varied between nominally identical sensors, from about 0.31 pF per % MC (Sensor A) and 0.46 pF per % MC (Sensor B) to as high as 0.84 pF per % MC across all four fabricated sensors. This supports per-sensor calibration but means the sensors are not interchangeable.

Future calibration should use more points per sensor across the full target range, include repeated runs, and be validated on data not used for fitting, ideally with a dynamic run referenced against gravimetric MC.

6.2 Physical Degradation and the Case for Encapsulation

Repeated wetting caused visible degradation of the graphene traces (Figure 10), and the anomalous behaviour of the two excluded sensors is consistent with this. Liquid bridging across the interdigital fingers also appeared to set the upper limit of the calibration, since readings became unreliable above approximately 20% MC, once liquid water could span the 0.5 mm electrode gaps. This is consistent with the transient dip shown by one excluded sensor between $t = 0.5$ and $t = 1.2$ hours, where temporary bridging would disturb the fringing field the sensor depends on. The flattened response of the other excluded sensor may instead reflect a degraded electrical contact at the header-pin interface. Neither mechanism was confirmed. The flat-topped shape of the first sensor's response before the dip could also indicate measurement saturation or a temporary connection loss, so inspecting the raw capacitance data and checking continuity and resistance across the sensor pins would help distinguish these explanations.

The partial compensation obtained by adjusting the intercept with a drift coefficient has a further limitation. An intercept shift corrects a constant offset, but loss of sensitivity is a change in slope, which an intercept correction cannot fully recover.

That two of the four sensors were affected indicates that unencapsulated LIG is adequate for controlled laboratory validation but is not yet robust enough for field deployment. This reinforces the decision to exclude environmental hardening from the project scope and points to encapsulation as a necessary next step before any in-situ trial. A vapour-permeable layer over the electrodes should stop surface water bridging the gaps, which could extend the usable range towards the fibre saturation point; wider finger spacing is another option. Any encapsulant would alter the dielectric environment of the fringing field, so sensors would need to be recalibrated after encapsulation.

6.3 Comparison with Existing Monitoring Approaches

Relative to the pin-type conductance sensors that dominate current practice (Section 2.3), the LIG-capacitive approach allows continuous, non-invasive monitoring without puncturing struc-

tural members. It also avoids the signal generators and machine-learning classification required by PZT-based active sensing (Section 2.4). Compared with the hygrometric method of Imamura et al. [4], LIG sensing removes the need for machined cavities and sealed void spaces, at the cost of the durability problems described above. LIG capacitive sensing is therefore a potentially lower-cost and more scalable alternative, though currently a less weatherproof one.

6.4 Limitations

The conclusions of this study are limited by the following factors:

- **Sensor selection:** two of the four sensors were excluded after the trial, so the reported sensors were selected partly on the basis of their dynamic behaviour. With only two sensors reported.
- **Calibration:** a small number of calibration points per sensor, with none between oven-dry and approximately 11% MC (which could not be conditioned at ambient humidity) or above approximately 20% MC (where liquid bridging occurred), and no held-out validation data.
- **Reference measurement:** reference moisture contents came from a low-cost resistive pin meter with no manufacturer-stated accuracy, using a single reading per sample at shallow pin depth, which reflects near-surface moisture. Its error is therefore unknown and is included in the scatter of the calibration data. During the dynamic trial the reference was available only as the 10% MC starting value, so accuracy over the drying cycle was not verified.
- **Trial scope:** a single 8-hour drying trial following surface wetting, which does not assess long-term drift, repeatability across wetting cycles, or response to moisture gradients through the thickness of the timber. The sensing depth of the electrode geometry was also not characterised.
- **Absence of encapsulation:** damage to two of the four sensors confounds loss of sensitivity with genuine moisture response in those channels, and the retained sensors may be similarly affected to a lesser degree.
- **Specimen type:** testing used solid-sawn Radiata Pine rather than bonded Cross-Laminated Timber so that the results reflect the behaviour of a sensor in a single lamella, but not the behaviour of one measuring the mass timber assembly as a whole. Theoretically, multiple sensors can be distributed throughout the assembly on different lamellae to get a complete map of the moisture; further research could apply this.

7 Conclusion

This project demonstrated a low-cost, non-destructive, in-situ moisture sensing system for New Zealand timber, integrating Laser-Induced Graphene (LIG) interdigitated capacitive sensors with an ESP32-C3 microcontroller and a custom Python data acquisition platform. Four LIG sensors were fabricated directly onto Radiata Pine substrates, calibrated against known moisture contents, and validated in a continuous 8-hour drying trial following controlled surface wetting.

The system successfully met its core objectives: it characterised sensor response across the target moisture range, converted raw capacitance to calibrated Moisture Content values in real time, and transmitted this data wirelessly for live and historical visualisation. Sensors A and B demonstrated ideal exponential desorption behaviour consistent with expected timber drying physics, confirming the underlying sensing principle is sound.

At the same time, the results exposed the central limitation of the current design: unencapsulated LIG traces are vulnerable to physical degradation from direct water contact, producing anomalous readings in two of the four channels and complicating calibration reliability. This finding, rather than undermining the approach, clarifies the next necessary step for translating this work toward practical deployment.

Future work should prioritise:

- vapour-permeable encapsulation of the LIG traces to protect against mechanical and liquid-induced degradation while preserving moisture sensitivity.
- expanded calibration datasets spanning a wider moisture range with more replicates per sensor, to produce more reliable regression fits.
- longer-duration and multi-cycle wetting/drying trials to assess sensor drift and long-term stability. Beyond these refinements, extending testing to bonded Cross-Laminated Timber elements would begin to bridge the gap between this laboratory proof-of-concept and the full-scale, in-situ structural health monitoring system envisioned for the New Zealand mass timber industry.

Overall, there are definite benefits to this system, and situations in which it could be more desirable than currently available methods of timber moisture measurement.

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A Theoretical Capacitance of the IDT Sensor

This appendix shows the calculation of the theoretical capacitance of the selected interdigital transducer (IDT) geometry (Figure 2) using Hilberg's explicit formulation [10], as introduced in Section 3.2.

A.1 Geometry and Constants

The selected design has the parameters given in Table 2:

$$\begin{aligned} N &= 5 \text{ finger pairs} \\ w &= 1 \text{ mm} = 1 \times 10^{-3} \text{ m} \quad \text{(finger width)} \\ s &= 0.5 \text{ mm} = 5 \times 10^{-4} \text{ m} \quad \text{(finger spacing)} \\ l &= 5 \text{ mm} = 5 \times 10^{-3} \text{ m} \quad \text{(overlap length)} \end{aligned}$$

The relative permittivity of oven-dry timber was taken as $\epsilon_r = 2 \pm 0.5$ [9], the speed of light in vacuum as $v_0 = 2.998 \times 10^8$ m/s, and the constant $377\ \Omega$ is the impedance of free space.

A.2 Validity Condition

Hilberg's formulation applies when

$$0 < \frac{s}{s+2w} \leq \frac{1}{\sqrt{2}} \approx 0.707.$$

For this geometry,

$$s + 2w = 0.5 + 2(1) = 2.5 \text{ mm}, \qquad k = \frac{s}{s+2w} = \frac{0.5}{2.5} = 0.2,$$

which satisfies the condition.

A.3 Capacitance per Finger Pair

The capacitance per finger pair is

$$C = \frac{\epsilon_r \, l \, \ln\left(\dfrac{2\,(a+1)}{1-a}\right)}{377\pi v_0}, \qquad a = \sqrt[4]{1-k^2},$$

which is Equation 3 with the term inside the logarithm rewritten using

$$-\frac{2}{a-1}(a+1) = \frac{2(a+1)}{1-a}$$

.

Step 1: evaluate a.

$$a = \sqrt[4]{1-(0.2)^2} = \sqrt[4]{0.96} = 0.98985.$$

Step 2: evaluate the logarithm.

$$\ln\left(\frac{2\,(0.98985+1)}{1-0.98985}\right) = \ln\left(\frac{3.9797}{0.01015}\right) = \ln(392) = 5.971.$$

Step 3: evaluate the denominator.

$$377\pi v_0 = 377 \times \pi \times 2.998 \times 10^8 = 3.551 \times 10^{11}.$$

Step 4: substitute $\epsilon_r = 2$.

$$\begin{aligned} C_{\text{pair}} &= \frac{2 \times (5 \times 10^{-3}) \times 5.971}{3.551 \times 10^{11}} \\ &= 1.68 \times 10^{-13}\ \text{F} \\ &= 0.168\ \text{pF}. \end{aligned}$$

A.4 Uncertainty and Total Capacitance

The capacitance is proportional to ϵ_r, so the uncertainty in permittivity scales directly:

$$\Delta C_{\text{pair}} = 0.168 \times \frac{0.5}{2} = 0.042\ \text{pF}.$$

For the five finger pairs of the selected sensor,

$$C_{\text{total}} = N\,C_{\text{pair}} = 5 \times 0.168 = 0.841\ \text{pF}, \qquad \Delta C_{\text{total}} = 5 \times 0.042 \approx 0.2\ \text{pF}.$$

A.5 Comparison with Measurement

The theoretical total of 0.841 ± 0.2 pF is more than ten times smaller than the 10–12 pF measured from oven-dried samples. This calculation models only the capacitance between the electrode fingers in a uniform medium of permittivity ϵ_r. It does not include stray capacitance from the header pins, connection tracks and jumper wires, capacitance associated with the electrode backbones and the substrate-air boundary, or residual moisture in the timber, all of which add to the measured baseline.

B Excluded Sensors

Four sensors were fabricated and logged. Two showed anomalous behaviour and were excluded from the main results. Table 3 gives the correspondence between the labels used in the report and the logged data channels, and Figure 13 shows the excluded traces.

Table 3: Correspondence between report labels, logged channels and sensor status.

Report label	Logged channel	Original ID	Status
Sensor A	CIN2	B	Reported
Sensor B	CIN4	D	Reported
Excluded sensor 1	CIN1	A	Excluded
Excluded sensor 2	CIN3	C	Excluded

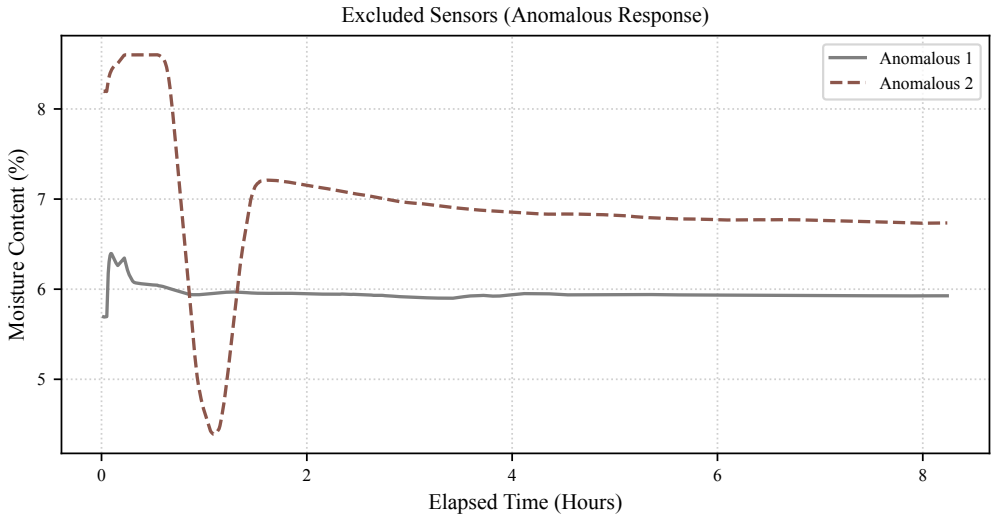


Figure 13: Moisture content traces of the two excluded sensors over the same 8-hour drying cycle.

Calibration curves and following equations are found in figure 14 and equations 6 and 7 respectively

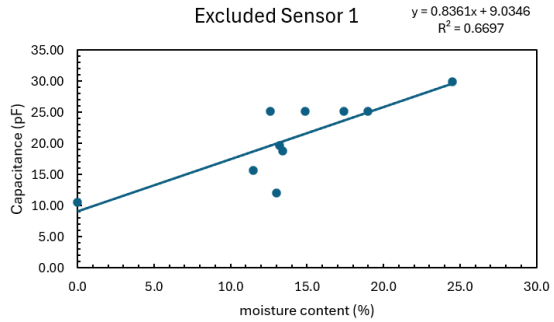
$$MC_1 = 0.8009C_1 - 2.6279 \tag{6}$$

$$MC_2 = 1.2706C_2 - 10.784 \tag{7}$$

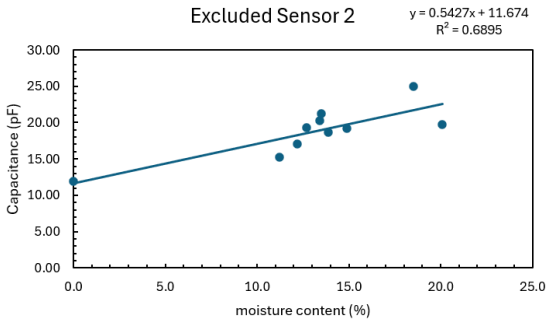
with a drift coefficients Defined a as:

$$\text{Drift Coefficient Excluded Sensor 1} = +6.5\ \%$$

$$\text{Drift Coefficient Excluded Sensor 2} = +6.7\ \%$$



(a) Excluded Sensor 1 ($y = 0.8361x + 9.0346$, $R^2 = 0.6697$)



(b) Excluded Sensor 2 ($y = 0.5427x + 11.674$, $R^2 = 0.6895$)

Figure 14: Experimental calibration curves plotting capacitance (pF) against moisture content (%) for Excluded Sensors 1 and 2.